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Project Name	AirAware AI — AI-Based Air Quality Information and Guidance System
Domain	Environmental Monitoring
Problem Statement	Malaysia provides air quality information through the Department of Environment (DOE) and the Air Pollutant Index Management System (APIMS). However, some users may find it difficult to understand Air Pollutant Index (API) readings, technical air pollution terms, and relevant precautions. Air quality information is also distributed across different government documents, requiring users to search multiple sources to obtain explanations.
Evidence of the Problem	Evidence 1: A study by Chin et al. (2019), involving 214 respondents from Klang Valley and Iskandar Malaysia, found that more than 60% of respondents had positive perceptions of their local air quality despite evidence of harmful pollution levels. This suggests that some Malaysians may underestimate air pollution risks, highlighting the importance of providing clear and understandable air quality information. Source / citation: Chin et al. (2019). *Public awareness and support for environmental protection—A focus on air pollution in peninsular Malaysia*. PLOS ONE. Evidence 2: A study involving 305 participants in Kuala Lumpur and surrounding areas found that individuals with greater knowledge about haze were more likely to demonstrate concern and engage in protective behaviours, such as checking air pollution readings and avoiding outdoor activities during haze episodes. This highlights the importance of accessible and accurate air pollution information to support informed decisions. . Source / citation: *The Link between Knowledge, Attitudes and Practices in Relation to Atmospheric Haze Pollution in Peninsular Malaysia* (2015). PLOS ONE.

Initial Related Work (at least 3)	Study / solution 1: Air Pollutant Index Management System (APIMS) Source / citation: Department of Environment Malaysia — APIMS Key limitation: APIMS provides air quality readings and classifications, but it is not primarily designed as a conversational assistant that explains information through natural-language questions. Study / solution 2: Department of Environment (DOE) Official Environmental Documents Source / citation: New Malaysia Ambient Air Quality Standard Key limitation: The documents provide authoritative technical information, but users may need to search and interpret the information manually. Study / solution 3: Traditional Keyword-Based Document Search Source / citation: Existing information retrieval approaches, such as TF-IDF and BM25 Key limitation: Keyword-based search may struggle to retrieve relevant information when users phrase questions differently from the terminology used in official documents.
Engineering Gap / Limitation	What practical or technical limitation remains unresolved? Gap / Limitation: Existing air quality information sources provide numerical readings and technical environmental information, but they may not offer a unified, natural-language interface that retrieves relevant passages and explains them with supporting references.
Initial Project Objectives (2–3)	O1: To design an AI-based air quality information system using official Malaysian environmental documents. O2: To develop a web-based AI assistant using semantic retrieval and RAG. O3 (if applicable): To evaluate retrieval relevance, answer accuracy, citation correctness, and response time.
Target Users	Malaysian residents, particularly students, outdoor workers, and members of the public who want to understand air pollution information and API readings.

Data Sources (at least 2)	Source 1: Malaysia Ambient Air Quality Standard — URL: _____ Source 2: Air Pollutant Index Information — URL: _____ Source 3 (if any): Clean Air Regulations 2014 — URL:
Similar to EduAdvisor	Both systems use Natural Language Processing and document retrieval to answer user questions based on official documents. Then, both aim to reduce the effort required to search through multiple information sources.
Different from EduAdvisor	EduAdvisor focuses on academic information such as course requirements and university policies. AirAware AI focuses on Malaysian air quality information, API interpretation, pollutant explanations, and general environmental precautions. Furthermore, AirAware AI also combines numerical API classification rules with AI-based semantic retrieval to explain air quality information.
Responsible AI Risks (at least 2)	Risk 1: Hallucination The AI may generate incorrect explanations or recommendations that are not supported by official environmental documents. Risk 2: Outdated Information The system may use outdated air quality guidelines or standards. Risk 3 (if any): Misleading Health Guidance The AI may provide overly confident advice about outdoor activities or individual health risks.